

Unit 8 – Inference for Categorical Data: Chi-Square
Notes 3 – Calculator Commands and the Follow-Up Analysis

Calculator Commands for Chi-Square

Both chi-square tests are located under
STAT → TESTS

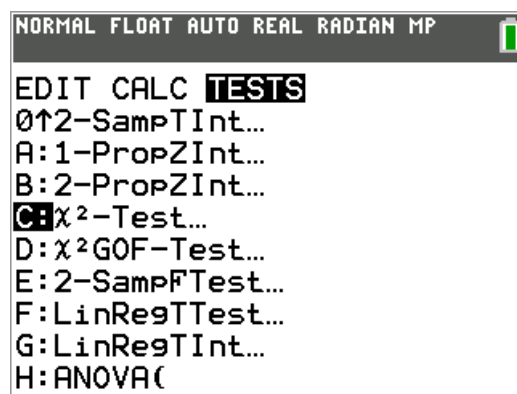
For the GOF Test:

- Observed Counts go in List 1
- Expected Counts go in List 2

For the homogeneity test:

- The observed counts go in a matrix: [A]
- The expected counts will get stored in the matrix listed next: [B] – you do not have to calculate this before!

*Make sure in your observed counts, you do not include the totals column or row



Example 1: The Mice from Notes 1

	Red	Brown	Pink	Blue
Observed	110	57	32	13
Expected	119.25	39.75	39.75	13.25

What are the chi-square statistic and p-value?

CNTRB stands for “contribution”, and if you scroll over, you will see each of the contributions to the chi-square statistic.

A single contribution is $\frac{(O-E)^2}{E}$ for a cell. This helps with the follow-up analysis and determining which proportion was the most different than expected.

Example 2: Living with the Parents from Notes 2

	With Parents	In Apartment	Dormitory	Total
Female	61	29	60	150
Male	60	27	113	200
Total	121	56	173	350

What are the chi-square statistic and p-value?

Follow Up Analysis

- If we reject the null hypothesis, a follow-up analysis helps determine which component of the distribution was the most different from what we expected.
- The computer output below is an example of a follow-up analysis for a chi-square test for homogeneity.

Chi-Square Test Follow-up:

	With Parents	In Apartment	Dormitory	Key: Observed Expected CNTRB
Female	61	29	60	
	51.857	24	74.143	
	1.612	1.042	2.698	
Male	60	27	113	
	69.143	32	98.857	
	1.209	0.781	2.023	

Chi-Sq = 9.365 DF = 2 P-Value = 0.00926

Looking at the output, we see that just two of the six components that make up the chi-square statistic contribute about 5 (almost 50%) of the total $\chi^2 = 9.365$.

Which Test?

Many studies involve comparing the proportion of successes for each of several populations or treatments.

- The _____ allows us to test the null hypothesis $H_0: p_1 = p_2$, where p_1 and p_2 are the actual proportions of successes for the two populations (or treatments)
- The _____ allows us to test $H_0: p_1 = p_2 = \dots = p_k$. This null hypothesis says that there is no difference in the proportions of successes for the k populations (or treatments). This is for a single categorical variable across many populations (or treatments).
- The _____ allows us to test the null hypothesis for multiple categorical variables across many populations (or treatments). Remember that this data is best displayed in a two-way table.

Example

A college professor at a large university conducts a study to determine which of the Big Five personality traits a person displays the most of. He took three separate random sample of college majors and recorded the personality trait of each of the individuals. The results are shown in the table below:

	Open	Conscientious	Extrovert	Agreeable	Neurotic	Total
Business	25	26	30	38	32	151
Engineering	38	27	21	29	36	151
Communication	24	25	39	28	22	138
Total	87	78	90	95	90	440

If a student is considered an extrovert, is there is difference in whether they choose a business major or engineering major?	Is there a difference in the distribution of personality types among business majors?	Is there is difference in the distribution of personality types across certain college majors?

a) Let's perform a chi-square test for homogeneity. Define the null and alternative hypotheses.

b) Using your calculator, find the expected counts in the table below.

Expected Counts	Open	Conscientious	Extrovert	Agreeable	Neurotic
Business					
Engineering					
Communication					

c) Are the conditions met to perform inference?

d) The table below shows each of the contributions for the chi-square statistic:

Contributions	Open	Conscientious	Extrovert	Agreeable	Neurotic
Business	0.79	0.022	0.025	0.894	0.04
Engineering	2.221	0.002	3.165	0.398	0.847
Communication	0.396	0.012	4.111	0.108	1.374

Using this information, find the chi-square statistic and the p-value for the chi-square test for homogeneity.

e) What are the largest contributions to the chi-square statistic and what does this mean in the context of the problem?